

SMART AUTOMATIC POWER FACTOR CORRECTION SYSTEM

1. Abstract

Power factor is an important parameter in an AC electrical system. A low power factor causes higher current, increased losses, voltage drop and inefficient utilization of electrical equipment. This project presents a **Smart Automatic Power Factor Correction (APFC) System** that monitors the electrical load and automatically switches capacitor banks according to the required reactive power compensation.

The system uses voltage and current sensing, a microcontroller, relay switching and capacitor banks. When the power factor decreases below the desired value, the controller activates suitable capacitors. When the power factor improves, unnecessary capacitors are disconnected. This provides automatic power factor improvement and demonstrates the practical application of power-system and embedded-control concepts.

Keywords: Power Factor, APFC, Capacitor Bank, Reactive Power, Arduino, Relay, Current Transformer.

2. Introduction

In AC electrical systems, many industrial and domestic loads such as induction motors, transformers and fluorescent lamps require reactive power for their operation. These loads generally have an inductive nature and therefore consume both active power and reactive power.

Power factor is the ratio of active power to apparent power.

Power Factor = Active Power / Apparent Power

A low power factor results in increased current for the same useful power output. Higher current causes additional losses in cables and electrical equipment.

One common method of improving power factor is to connect capacitors in parallel with the inductive load. In an automatic power factor correction system, capacitors are switched automatically according to the load condition.

3. Problem Statement

Electrical loads do not always operate at a constant power factor. Manual capacitor switching may not provide suitable compensation when the load changes.

The main problems associated with low power factor are:

- Increased line current
- Higher I^2R losses
- Voltage drop
- Poor utilization of electrical equipment
- Increased apparent power demand
- Reduced overall system efficiency

Therefore, an automatic system is required to monitor the load condition and switch the required capacitor stages automatically.

4. Objectives

The main objectives of this project are:

1. To monitor the electrical parameters of an AC load.
2. To determine the power factor of the load.
3. To detect low power-factor conditions.
4. To automatically switch capacitor banks.

5. To improve the power factor of the electrical system.
6. To reduce unnecessary reactive power demand.
7. To understand practical power-factor correction techniques.

5. Basic Theory of Power Factor

In an AC circuit:

$$\text{Power Factor} = \cos \phi$$

where:

- ϕ = phase angle between voltage and current.

The three important power quantities are:

Active Power

Active power is the useful power consumed by the load.

$$P = VI \cos \phi$$

Unit: Watt (W)

Reactive Power

Reactive power is associated with energy storage in inductors and capacitors.

$$Q = VI \sin \phi$$

Unit: VAR

Apparent Power

Apparent power is the product of RMS voltage and RMS current.

$$S = VI$$

Unit: VA

The relationship is:

$$S^2 = P^2 + Q^2$$

6. Power Factor Correction

For an inductive load, current lags the voltage. A capacitor produces leading reactive current. When a capacitor is connected in parallel with the inductive load, the reactive power supplied by the capacitor compensates part of the inductive reactive power.

The required capacitor reactive power can be calculated using:

$$Q_c = P(\tan \phi_1 - \tan \phi_2)$$

where:

- Q_c = required capacitor reactive power
- P = active power
- ϕ_1 = initial phase angle
- ϕ_2 = desired final phase angle

By selecting suitable capacitor stages, the system can automatically maintain the power factor near the desired value.

7. Existing System

In a conventional manual power-factor correction system, capacitors are switched manually based on the estimated load condition.

Disadvantages

- Manual operation is required.
- Compensation may not match changing load conditions.
- Over-compensation can occur.
- Continuous monitoring is difficult.
- It is less suitable for rapidly changing loads.

8. Proposed System

The proposed system is an automatic power factor correction system.

The controller receives information from voltage and current sensing circuits. It determines the electrical condition of the load and controls relay outputs connected to different capacitor stages.

Basic operation:

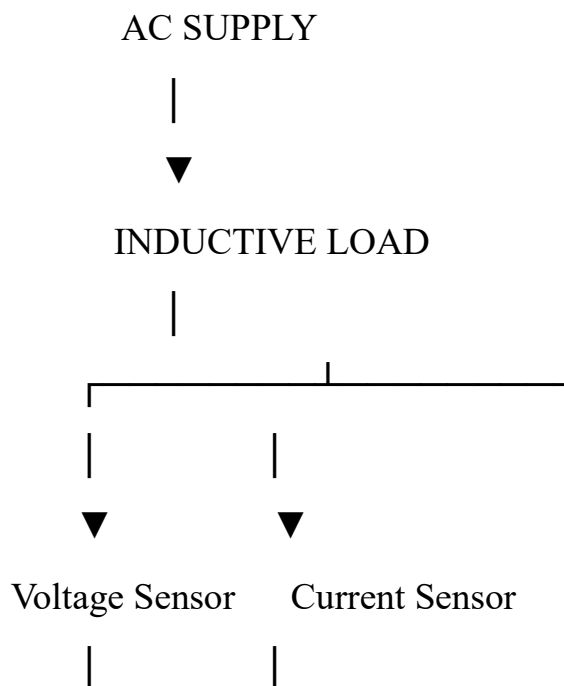
Load → Voltage/Current Sensing → Microcontroller → Relay Driver → Capacitor Bank

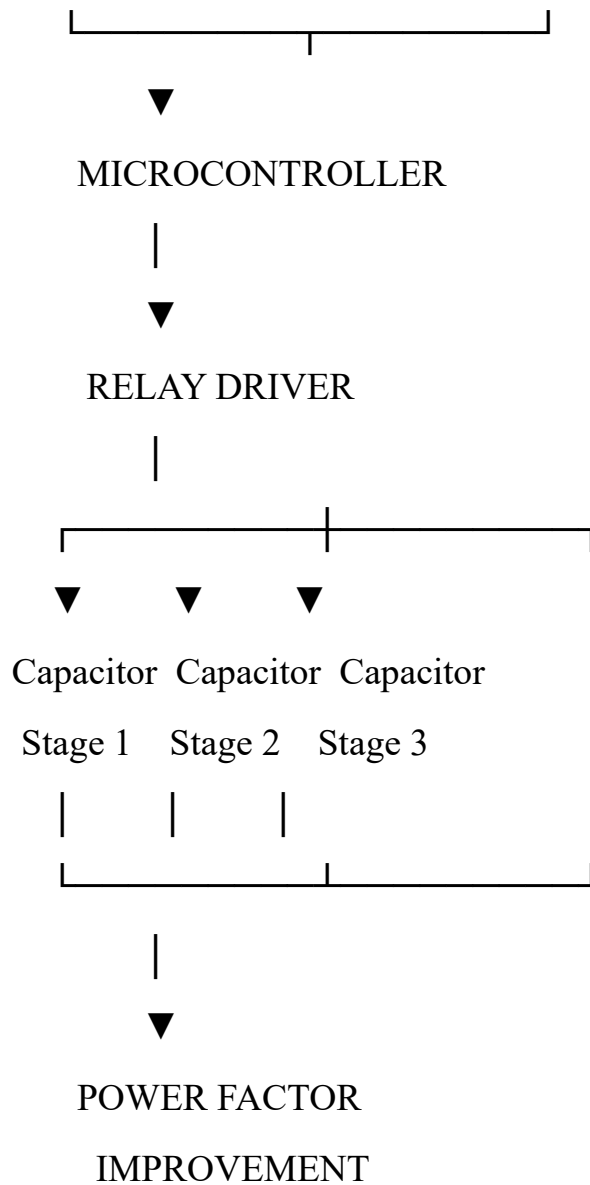
When the measured power factor is low, the controller connects an appropriate capacitor stage.

When the power factor reaches the desired range, the controller stops adding capacitors.

If excessive compensation occurs, the controller disconnects a capacitor stage.

9. Block Diagram





10. Components Required

Component	Purpose
Arduino/ESP32	Main controller
Current Transformer	Current measurement
Voltage Sensor	Voltage measurement

Component	Purpose
Relay Module	Capacitor switching
Capacitor Bank	Reactive power compensation
LCD Display	Parameter display
Resistors	Sensing/protection circuits
Power Supply	Provides control power
Inductive Load	Test load
Connecting Wires	Circuit connections

11. Working Principle

The working process is as follows:

Step 1 – AC Supply

The AC supply is connected to the electrical load.

Step 2 – Load Produces Reactive Power

If the load is inductive, the current lags the voltage and the power factor becomes less than unity.

Step 3 – Parameter Sensing

Voltage and current sensors measure the electrical parameters.

Step 4 – Controller Processing

The microcontroller receives the sensor signals and determines the power factor condition.

Step 5 – Decision

The controller compares the measured power factor with the target value.

For example:

- PF below target → capacitor required
- PF near target → maintain present condition
- PF excessively leading → disconnect capacitor

Step 6 – Capacitor Switching

The controller activates the appropriate relay.

The relay connects the required capacitor stage to the load.

Step 7 – Power Factor Improvement

The capacitor supplies leading reactive power and compensates the inductive reactive power.

Step 8 – Continuous Monitoring

The controller continuously monitors the load and changes the capacitor stages when required.

12. Control Algorithm

START



Initialize sensors and relays



Measure voltage and current



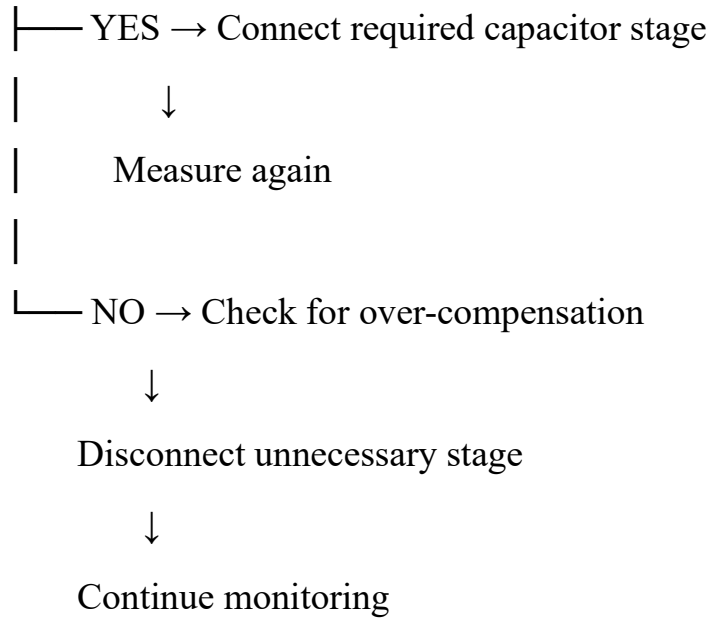
Calculate/estimate power factor



Compare PF with target value



Is PF below target?



13. Example of Automatic Control

Assume the target power factor is approximately **0.95**.

Condition	Controller Action
PF = 0.70	Connect capacitor Stage 1
PF = 0.82	Connect additional stage if required
PF = 0.94	Maintain condition
PF = 0.97	Maintain/disconnect depending on control limits
Leading PF	Disconnect capacitor stage

The actual switching thresholds depend on the selected hardware and control algorithm.

14. Circuit Description

The current transformer is used to sense the load current. A suitable voltage-sensing circuit provides the voltage information to the controller.

The microcontroller processes the sensor signals. Its output pins are connected to relay-driver inputs.

The relay contacts control different capacitor stages.

The capacitor bank is connected in parallel with the inductive load through appropriate switching and protection components.

Note: Practical AC mains circuits require proper isolation, fusing, contactors/relays rated for the application, discharge resistors and electrical safety precautions. A low-voltage prototype should be used for student demonstration where possible.

15. Software / Programming

The microcontroller program performs the following functions:

1. Initialize input and output pins.
2. Read voltage and current signals.
3. Process the measured values.
4. Determine the power-factor condition.
5. Compare the result with predefined thresholds.
6. Control the relay outputs.
7. Display the system status.
8. Repeat the monitoring process continuously.

Simplified Pseudocode

START

Initialize sensors

Initialize relay outputs

Initialize display

WHILE system is ON:

Read voltage

Read current

Determine power factor

IF power factor < lower limit:

Connect capacitor stage

ELSE IF power factor > upper limit:

Disconnect capacitor stage

ELSE:

Maintain present capacitor stage

Display voltage, current and PF

END WHILE

16. Expected Results

The expected result of the proposed system is improvement in the power factor of an inductive load.

For example:

Parameter	Before Correction	After Correction
Power Factor	0.70	Around 0.90–0.98
Reactive Power	Higher	Reduced
Line Current	Higher	Reduced
System Efficiency	Lower	Improved

These values are illustrative. Actual experimental values depend on the load, capacitor ratings, sensing method and control settings.

17. Advantages

- Automatic capacitor switching
- Improved power factor
- Reduced reactive power demand
- Reduced line current
- Reduced electrical losses
- Better utilization of electrical equipment
- Continuous monitoring
- Suitable for variable loads
- Demonstrates practical embedded control

18. Limitations

- Accurate sensing is required.
- Capacitor selection must match the load.
- Relay/contactors must be properly rated.
- Harmonic currents can affect capacitor-bank operation.

- Incorrect switching can cause over-compensation.
- AC mains implementation requires proper electrical protection.

19. Applications

The system can be applied in:

- Small industries
- Manufacturing plants
- Motor-driven systems
- Workshops
- Commercial buildings
- Pumping systems
- Industrial distribution panels
- Educational electrical laboratories

20. Future Scope

The project can be further improved by adding:

1. IoT-based remote monitoring.
2. Mobile application support.
3. Automatic energy-data logging.
4. Cloud-based monitoring.
5. More accurate power-factor measurement.
6. Automatic capacitor-stage optimization.
7. Harmonic monitoring.
8. Three-phase power-factor correction.

9. Industrial communication such as Modbus.
10. Dashboard-based real-time monitoring.

21. Conclusion

The **Smart Automatic Power Factor Correction System** provides an effective method for monitoring and improving the power factor of inductive electrical loads.

The system uses sensing circuits, a microcontroller, relay switching and capacitor banks to provide automatic reactive-power compensation. By automatically selecting capacitor stages according to the load condition, the system can reduce unnecessary reactive power demand and improve the utilization of electrical equipment.

This project provides practical knowledge of **power factor, reactive power, capacitor banks, electrical measurement, relay control and embedded systems**, making it a useful educational project for Electrical and Electronics Engineering students.

22. References

1. Electrical Machines and Power System Engineering textbooks.
2. Power Factor Correction application notes from electrical-component manufacturers.
3. Arduino technical documentation.
4. ESP32 technical documentation.
5. Standard electrical engineering references on AC circuits and reactive power.
6. Capacitor-bank manufacturer application guides.

23. GitHub Project Information

Repository Name:

Smart-Automatic-Power-Factor-Correction

Suggested Repository Description:

An educational automatic power factor correction system that monitors electrical load conditions and controls capacitor stages to improve power factor.

Suggested Topics:

electrical-engineering

power-factor

apfc

power-systems

arduino

embedded-systems

capacitor-bank

eee-project

24. Suggested GitHub Folder Structure

Smart-Automatic-Power-Factor-Correction/

|

|— README.md

|

|— Report/

| └— Smart_Automatic_Power_Factor_Correction_Report.pdf

|

|— Circuit/

| └— circuit_diagram.png

|

|— Code/

- | └─ pf_correction.ino
- |
- |─ Images/
- | └─ project_setup.jpg
- |
- |─ Results/
- └─ experimental_results.xlsx